

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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Anonymous submission

ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling representations from a strong, pretrained vision encoder into intermediate diffusion features. We identify and answer a central question for this paradigm: which property of the target representation actually matters for generation, its *global* semantic information (e.g., measured by ImageNet-1K accuracy) or its spatial structure (i.e. pairwise cosine similarity between patch tokens)? The prevailing view treats stronger global semantic performance as the decisive indicator of a better target representation. We test this view through a large-scale empirical analysis spanning 27 vision encoders and multiple model scales. The evidence points decisively in the opposite direction: spatial structure, rather than global performance, is the dominant predictor of generation performance under representation alignment. Building on this finding, we introduce two direct modifications that explicitly strengthen the transfer of *spatial* information. We replace the standard MLP projection layer in REPA with a simple convolution layer and add a spatial normalization layer for the external representation. The resulting method, iREPA, requires fewer than four lines of code and consistently improves the convergence speed of REPA across diverse vision encoders, model sizes, and training variants (including REPA, REPA-E, Meanflow, JiT, and others). Our results establish spatial structure as a key mechanism in representational alignment and provide a practical route to more effective training of generative models.

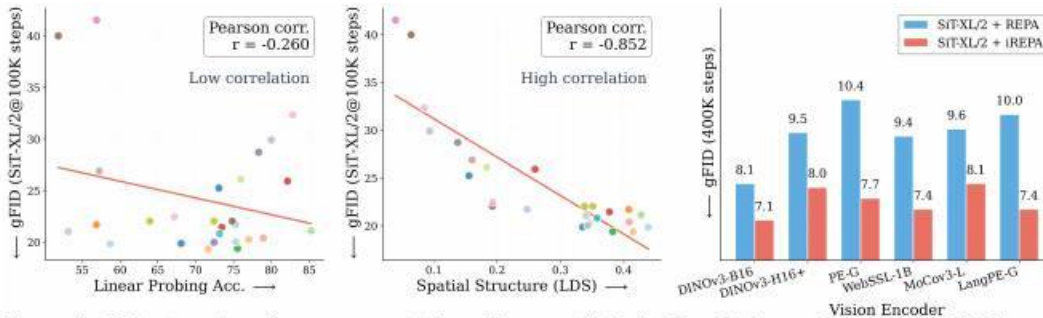


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders shows that spatial structure, not global performance measured by linear probing accuracy, is the stronger predictor of target-representation performance for generation. *Right:* We turn this mechanism into two simple modifications that accentuate spatial-feature transfer from the target representation to the diffusion model. The resulting method consistently accelerates REPA across diverse settings.

1 INTRODUCTION

Representation alignment has become a highly effective tool for accelerating diffusion transformer training (Ma et al., 2024; Peebles & Xie, 2023). By aligning internal diffusion representations with pretrained self-supervised visual encoders, recent methods deliver clear gains in convergence speed and final generation performance (Yu et al., 2024; Leng et al., 2025a). Yet the mechanism behind these



Figure 2: **Motivating examples — spatial structure matters.** These comparisons expose a clear mismatch between ImageNet accuracy and generation quality. *Left:* PE-G has much higher validation accuracy (82.8% vs. 53.1%) but worse generation quality than SpatialPE-B (Bolya et al., 2025). *Right:* WebSSL-1B (Fan et al., 2025) likewise has stronger global performance (76.0% vs. 53.1%) but worse generation. **Spatial Self-Similarity:** Spatial structure provides the stronger signal for generation quality; §3 defines the metric. All results reported at 100K using SiT-XL/2 and REPA.

gains has remained unresolved: self-supervised encoders provide both global semantic information and structured patch-level geometry, and the field has not established which property drives the improvement. This paper isolates that mechanism by asking whether REPA benefits primarily from stronger global semantic information, commonly measured by linear probing performance, or from spatial structure, captured by relationships among patch token representations.

Resolving this question is essential because it determines how target representations should be selected and how their signal should be transferred. The dominant assumption is that encoder quality for representation alignment should track ImageNet-1K validation accuracy, a standard proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this view, target representations with stronger ImageNet performance should yield better generation (Yu et al., 2024). Consequently, improvements in diffusion-feature linear probing accuracy are often treated as direct evidence that representation alignment works by importing global semantic information. The following quote from REPA (Yu et al., 2024) captures this prevailing interpretation:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we directly challenge this interpretation and show that it is incomplete. We systematically compare global semantic information and spatial structure across a broad set of vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024). This analysis yields three clear findings that sharpen the current understanding of representation alignment.

Higher validation accuracy does not imply better representation for generation. Contrary to the standard expectation, encoders with stronger global semantic performance, measured by ImageNet-1K linear probing accuracy, often produce worse generations. For example, PE-Spatial-B is a spatially tuned model derived from PE-Core-G (Bolya et al., 2025). Although PE-Spatial-B has much lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), it yields better generation with REPA (Figure 2). WebSSL-1B (Fan et al., 2025) provides a second example: despite substantially higher global performance (76.0% vs. 53.1%), it produces worse generation. These comparisons demonstrate that ImageNet accuracy is not a reliable proxy for target-representation quality in REPA; in Section §2, we show that this pattern extends across target representations and points to a core principle in how representation alignment benefits diffusion training.

Spatial structure strongly predicts generation performance. We introduce simple spatial self-similarity metrics, rooted in local self-similarity analysis (Shechtman & Irani, 2007), that quantify how patch-token similarity changes with spatial distance (§3). Across 27 encoders, these spatial metrics correlate far more strongly with gFID than linear probing accuracy does, and the same spatial explanation accounts for the controlled trends in Figure 3. This result reframes the evidence: the features that matter for generation are not primarily the globally semantic features emphasized by

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

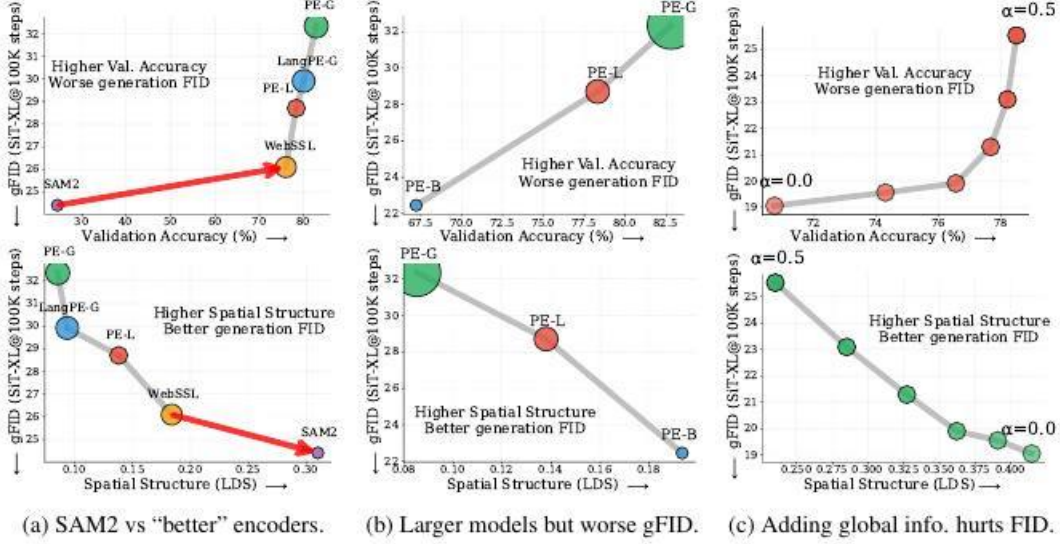


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Multiple controlled comparisons show that global performance is not a reliable predictor of generation FID under REPA. (a) SAM2-S, despite only 24.7% validation accuracy, achieves better generation performance than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can improve validation accuracy while degrading generation performance. (c) Adding global information to patch tokens via the CLS token improves global performance but hurts generation. **Bottom row:** Spatial structure, rather than global performance, provides the better indicator for generation; §3 gives the large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

classification accuracy, but the spatial organization preserved in the patch-token representation. Even classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005) provide useful REPA signal when they preserve this structure (§3).

Accentuating spatial transfer improves REPA. The spatial-structure finding also leads to an immediate practical improvement. We introduce iREPA, a minimal modification of REPA that strengthens the transfer of spatial information from the target representation to the diffusion features. iREPA replaces the MLP projection layer with a convolutional projection and applies spatial normalization to the target representation. Despite its simplicity, iREPA consistently accelerates convergence across target encoders, model sizes, alignment depths, and training recipes, including REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025); Figure 6 shows how the modifications preserve spatial structure during transfer.

Our contributions are:

- We show that higher global semantic performance of a target representation, as measured by ImageNet-1K accuracy, does not imply better generation performance with representation alignment. This directly revises the common selection criterion for target encoders.
- We introduce Spatial Structure Metrics (SSM) and demonstrate that spatial structure is a substantially stronger predictor of downstream FID performance than linear probing scores.
- We propose simple modifications that explicitly accentuate spatial-information transfer from the target representation to diffusion features. The resulting method improves convergence over REPA across target representations, model architectures, and training recipe variants (REPA, REPA-E, Meanflow w/ REPA, JiT w/ REPA, etc.).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with motivating evidence that stronger global performance, measured by ImageNet-1K accuracy, does not reliably translate into better generation performance with REPA. We then show in §3 that these otherwise puzzling results are consistently explained by the spatial structure of the target representation.

Recent vision encoders. We first compare recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025), makes the mismatch clear. As shown in Figure 2, PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%) yet performs worse as the REPA target representation (FID 32.3 vs 21.0). WebSSL-1B (1.2B) follows the same pattern: it achieves much higher ImageNet-1K accuracy (76.0% vs 53.1%) but produces worse generation when used as the target representation (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. We next stress-test the role of global information using SAM2-S (Ravi et al., 2024), a small 46M-parameter vision encoder with very limited global information and only 24.1% validation accuracy (Fig. 3a). Despite this weak global score, SAM2-S achieves better FID under REPA than encoders with dramatically higher ImageNet-1K accuracy, including PE-Core-G (82.8%). This result sharply separates global classification strength from usefulness as a REPA target.

Larger models within same encoder family can have similar or worse generation performance. The same disconnect appears within encoder families. Larger variants typically have stronger ImageNet-1K accuracy and are therefore expected to provide better target representations. For representation alignment, however, larger variants often yield similar (DINOv2) or even *worse* (PE, Cradio) generation performance (Fig. 3b). A related observation for DINOv2 was reported in (Yu et al., 2024) and attributed to all DINOv2 models being distilled from DINOv2-g and therefore sharing similar representations. In §3, we show that spatial structure provides a more direct explanation for this trend.

Adding more global information can hurt generation. We finally perform a controlled intervention that injects global semantics into local patch tokens using the CLS token, with DINOv2 as the encoder. The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As shown in Figure 3c, increasing α from 0 to 0.5 improves linear probing accuracy monotonically from 70.7% to 78.5%. Generation quality moves in the opposite direction: FID worsens from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

Together, these observations establish that global performance is a poor indicator of REPA performance. We next show in §3 that spatial structure provides the missing predictive signal.

- **Finding 1.** *Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.*

3 SPATIAL STRUCTURE MATTERS MORE

We posit that spatial structure, rather than global information, is the principal factor governing how useful a target representation is for generation. To test this claim, we define a set of straightforward metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then show that these spatial metrics correlate far more strongly with gFID than ImageNet-1K accuracy and explain the anomalous trends identified in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- **Self-similarity.** Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extracted patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- **Spatial distance.** Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ denote the spatial location of each of the T tokens in coordinate space, and let $d(\cdot, \cdot) \in \mathbb{N}$ denote the Manhattan distance between token pairs.

We quantify *spatial* self-similarity by measuring how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Larger values indicate stronger spatial organization: nearby patches remain more similar to one another than distant patches. By default, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

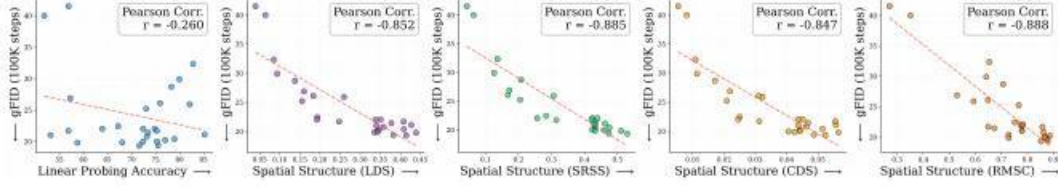


Figure 4: Spatial structure is a much stronger correlate of generation quality than linear probing. Across 27 diverse vision encoders with SiT-XL/2 and REPA, linear probing has weak correlation with FID (Pearson $|r| = 0.260$). In contrast, all spatial structure metrics are strongly correlated with generation performance: LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$). See Fig. 10 for detailed plots with encoder labels.

The final spatial self-similarity score (LDS) is the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, and the observed correlations remain robust to these choices. Appendix B provides exact details and alternative spatial metrics (CDS, SRSS, RMSC), which also perform effectively (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing. We conduct a large-scale correlation analysis across 27 diverse vision encoders. Figure 4 shows a clear separation: the commonly used linear probing metric has weak correlation with generation performance (Pearson $|r| = 0.26$), whereas every SSM metric exhibits much stronger correlation (Pearson $|r| > 0.85$).

Generalization across model scales. We further verify this relationship across model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing remains weak across scales ($|r| < 0.306$), while spatial structure remains strongly predictive of generation performance ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure explains previously unexplained trends. As shown in §2, ImageNet-1K accuracy fails as a predictive measure for representation alignment. The spatial metrics introduced above provide a consistent explanation for these failures.

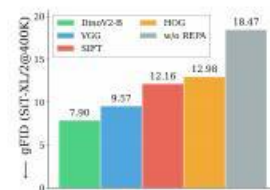
(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%) but performs worse as the REPA target representation (FID 32.3 vs 22.0). The spatial metric resolves this discrepancy: as shown in Figure 2, despite lower global performance, PE-Spatial-B preserves substantially stronger spatial structure than PE-Core-G, matching its better generation performance.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that SAM2 has much lower ImageNet-1K accuracy (24.1%) yet outperforms encoders with roughly $\sim 60\%$ higher accuracy. Fig. 3a shows that this advantage is directly explained by SAM2’s stronger spatial structure.

(3) *Larger models in same encoder family underperform:* As shown in Fig. 3b, larger models in the same encoder family can improve ImageNet-1K accuracy while degrading spatial structure, which in turn leads to worse generation performance under REPA. This interpretation also accounts for the observation in Yu et al. (2024) that larger DINOv2 variants can produce similar or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation:* Figure 3c shows that CLS-token mixing increases global performance but reduces spatial contrast among patch tokens. The mixed patch tokens become overly similar to otherwise unrelated tokens (e.g., foreground object to background), weakening spatial structure and degrading generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Yes. Classical spatial features such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) are not the strongest targets, but they still yield REPA gains. This result provides further evidence that representation alignment can benefit from spatial structure even without additional global semantic information.



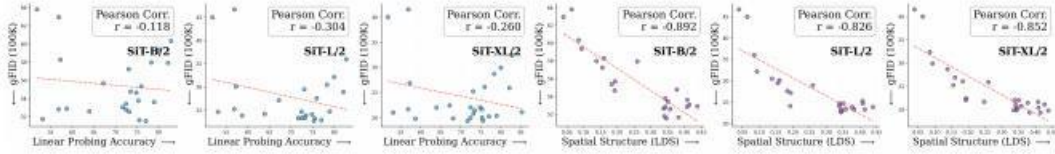
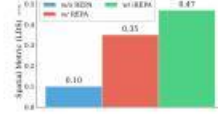


Figure 5: **Correlation analysis across model scales.** Spatial structure (right) remains the stronger correlate of gFID across model scales, whereas linear probing (left) remains weakly predictive.

Can spatial metrics explain gains from REPA itself? Yes. The proposed spatial metrics explain both the gains obtained by REPA and the additional gains from the improved training recipe, iREPA, introduced next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

The preceding analysis identifies spatial structure as the key signal that REPA should transfer. We therefore introduce two minimal modifications to the original REPA training recipe that deliberately preserve and amplify spatial features as they pass from the teacher vision encoder to the student diffusion transformer.

Convolutional projection layer instead of MLP. Standard REPA uses a 3-layer MLP projection to map diffusion features into the dimensionality of the external representation. Figure 6a shows that this projection is lossy for spatial structure and reduces spatial contrast between patch tokens. We replace it with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid. This convolutional projection preserves local relationships through its spatial inductive bias while keeping the modification simple.

Spatial normalization layer. Consistent with Siméoni et al. (2025), we find that patch tokens from pretrained vision encoders contain a substantial global component, as reflected by high linear probing scores for the mean of patch tokens (Figure 14). Figure 6b further shows that mixing this global information (mean of patch tokens) into the patch tokens improves global performance but reduces spatial contrast between individual patch tokens. The result is a less discriminative spatial map in which unrelated tokens, such as foreground and background patches, become overly similar.

Given the results in §3, we intentionally trade some global information (mean of patch tokens) for stronger spatial contrast among patch tokens. This sharper contrast provides the spatial signal that REPA needs: pairwise similarity between patch tokens. We implement this principle with a simple spatial normalization layer (Ulyanov et al., 2016) applied to the target-representation patch tokens:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We validate iREPA through extensive ImageNet 256×256 experiments. The evaluation is designed to test whether explicitly improving spatial transfer yields reliable gains beyond standard REPA:

- Does iREPA consistently accelerate diffusion-transformer convergence over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)

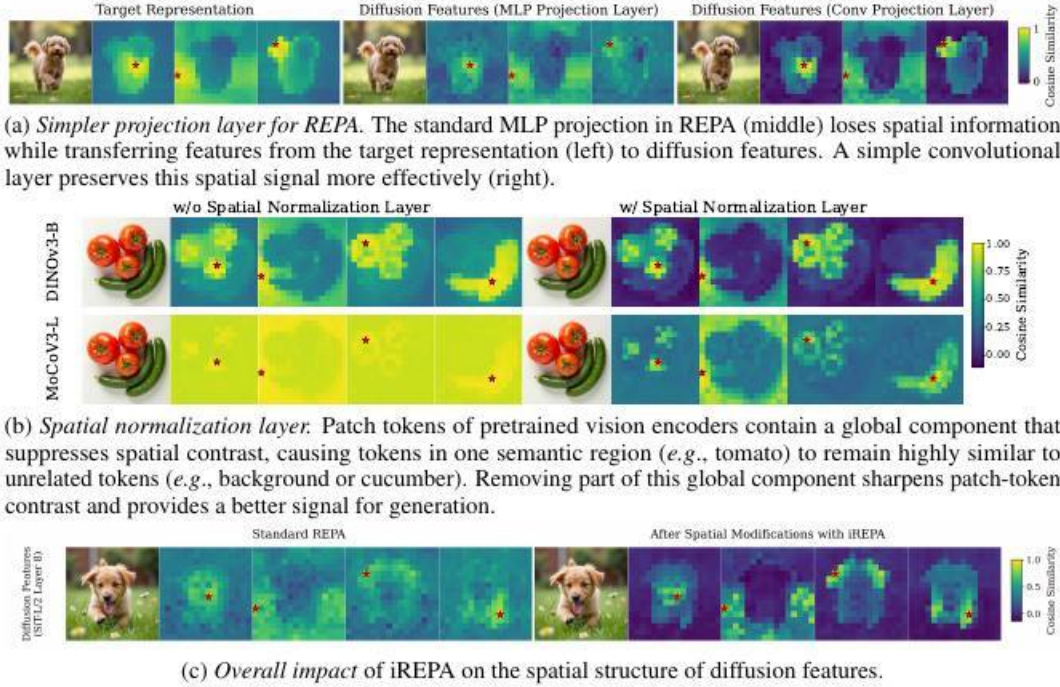


Figure 6: Two direct iREPA modifications strengthen spatial-feature transfer from the target representation to diffusion features. All results are reported with SiTL/2 with REPA at 100K.

- Does iREPA scale with model size and remain effective across training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA extend to newer representation-alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), as well as pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate iREPA across diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). Fig. 7 shows that iREPA consistently reaches better FID faster than baseline REPA across both target representations and model scales.

Target representation. We next test whether the improvement depends on a particular encoder. Fig. 8 and Table 4 show that iREPA improves generation quality across all evaluated vision encoders. Appendix C and Appendix E provide additional comparisons across all 27 encoders.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA delivers consistent gains over baseline REPA across encoder size, model scale, and alignment depth. Unless otherwise specified, all results use Dinov2 as encoder, SiTXL/2, 100K steps, and vanilla REPA as the baseline.

Encoder size. We evaluate iREPA across PE encoder sizes in Table 1a: PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance for every encoder size. The relative improvement also grows with encoder scale (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G), indicating that spatial transfer becomes increasingly valuable for larger target encoders.

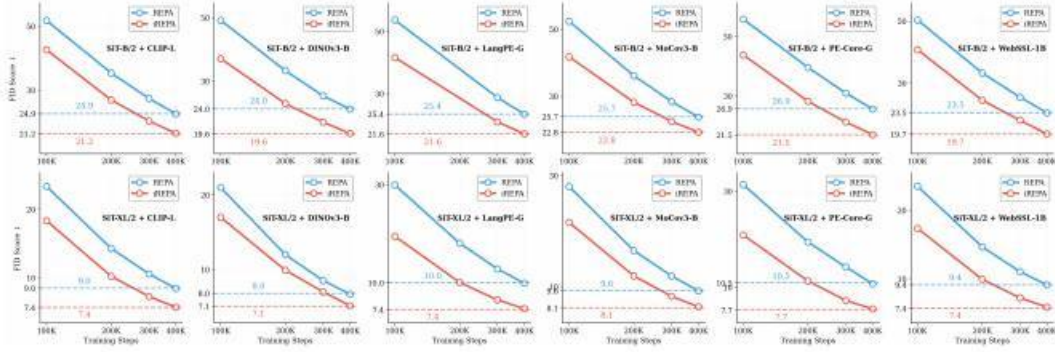


Figure 7: **Accentuating spatial features consistently improves convergence speed.** iREPA accelerates REPA across the evaluated encoders and model sizes by strengthening spatial transfer.

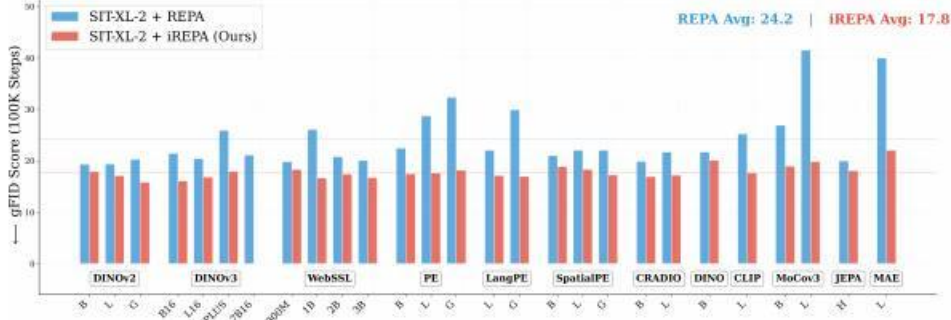


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating spatial-feature transfer from the target representation to diffusion features with iREPA consistently improves convergence speed. See Appendix C for more results across diverse settings.

Scalability. Table 1b evaluates iREPA across diffusion model scales. The spatial modifications improve performance at each scale, and the larger percentage gains for larger models show that iREPA scales with model capacity.

Encoder depth. Table 1c evaluates iREPA across alignment depths using SiT-B/2 at 100K iterations with DiNOv3-B as the target representation. iREPA improves over baseline REPA at every tested depth.

Ablation on different components. Table 2 isolates the two iREPA components. Both spatial normalization and the convolutional projection layer substantially improve generation quality over baseline REPA, and combining them gives the strongest results.

Training recipe. We further test iREPA beyond the original REPA recipe, including REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows that the same spatial improvements deliver convergence gains across these training recipes.

Classifier-free guidance. Table 4 evaluates iREPA with CFG across vision encoders at 400K training iterations. iREPA accelerates convergence both with and without classifier-free guidance, demonstrating that the spatial-transfer benefit is not tied to a particular sampling setting.

Pixel-space diffusion (JiT). We also evaluate iREPA on pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA with JiT (Li & He, 2025b); emphasizing spatial information consistently accelerates REPA across vision encoders (e.g., DiNOv2, DiNOv3, PE etc.).

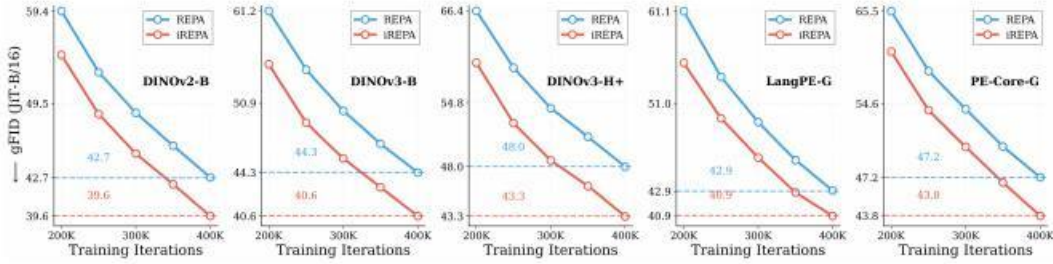


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves compare REPA and iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. Accentuating spatial-structure transfer consistently improves convergence speed for pixel-space diffusion models. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer Table 10 for further results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across encoders, both the spatial normalization layer and convolutional projection layer (§4) produce substantial convergence gains, and their combination gives the strongest results. All results are reported at 100K steps with SiT-XL/2.

- **Finding 3.** *Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).*

5 RELATED WORK

We summarize the most relevant work here and provide a more detailed discussion in Appendix H.

Representation alignment for generation. Recent work increasingly uses external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). In particular, REPA-style methods (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) show that aligning internal diffusion features with clean-image features from pretrained vision encoders can substantially improve convergence and generation quality. Related extensions apply the same principle to video generation and 3D generation (Zhang et al., 2025; Wu et al., 2025), while representation alignment also benefits unified models (Ma et al., 2025a). These successes establish representation alignment as an important training strategy, but they do not identify which property of the target representation drives the gain. Our work fills this gap by showing that spatial structure, rather than global semantic strength, is the more decisive factor for generation, and by translating that finding into an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent studies analyze the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). Continued self-supervised training can increase similarity between the global CLS token and patch tokens, which harms dense spatial tasks (Siméoni et al., 2025). Other work explicitly trains spatially tuned models for dense prediction (Bolya et al., 2025; Heinrich et al., 2024). We show that this global-spatial tradeoff is also central to generative training: for representation alignment, the spatial structure of a vision encoder matters more than its global information. This result provides a concrete criterion for selecting and designing external representations for generation.

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** The same spatial improvements transfer directly to REPA-E (a) and Meanflow with REPA (b), yielding consistent gains. For Meanflow, we report results at 1 and 4 NFEs. CFG value of 2.0 is used for Meanflow. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across different encoders with and w/o classifier-free guidance.** iREPA improves REPA across encoders both with and without CFG using SiT-XL/2 as the base model. See Appendix E for detailed SiT-B/2 and SiT-L/2 results across all encoders.

6 CONCLUSION

This paper identifies what drives the effectiveness of representation alignment: the spatial structure of the target representation, not its global information alone. Through large-scale empirical analysis, we show that ImageNet-style global performance is a weak and sometimes misleading indicator of REPA performance, while spatial self-similarity strongly predicts generation quality. We then turn this mechanism into iREPA, a simple training modification that accentuates spatial transfer through a convolutional projection layer and spatial normalization. iREPA consistently improves convergence over REPA across vision encoders, model scales, alignment depths, classifier-free guidance settings, and training recipes. These results establish spatial structure as a central design principle for representation alignment and provide a practical path toward more effective generative model training.

REPRODUCIBILITY STATEMENT

We provide the implementation details and hyperparameters needed to reproduce the experiments in Appendix G. We will release the code, model checkpoints, and analysis outputs upon acceptance.

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A DETAILED CORRELATION ANALYSIS

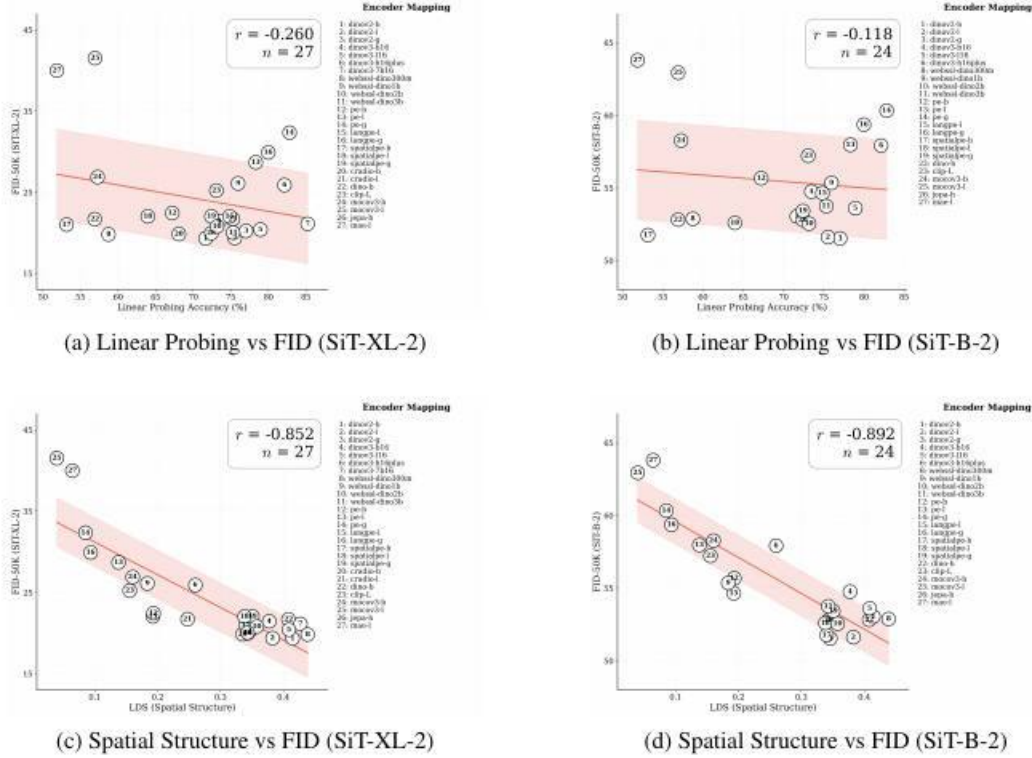


Figure 10: **Detailed correlation analysis with encoder labels.** Top row reports the correlation between linear probing accuracy and FID for SiT-XL-2 and SiT-B-2 models. Bottom row reports the corresponding correlation between spatial structure (LDS) and FID. Each point is labeled with its encoder quality number (see legend). Spatial structure is consistently the stronger correlate of generation quality across both model scales.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. This setup lets us measure spatial organization directly. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Larger values indicate stronger spatial organization: nearby patches remain more similar to one another than distant patches.

We next define several simple metrics that capture this spatial self-similarity structure in target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters. By default we use $r_{\text{near}} = r_{\text{far}} = H/2$, and the correlations remain robust to their exact values. Larger LDS values mean that, *on average*, nearby patches are more similar to each other than distant patches.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance and therefore stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsample it to $H \times W$. We then select anchor, positive, and negative points. Anchor and positive points come from inside the mask and should be semantically related, whereas the negative point comes from outside the mask and should be less related to the anchor. If an encoder preserves spatial structure, anchor-positive similarity should exceed anchor-negative similarity. We therefore define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

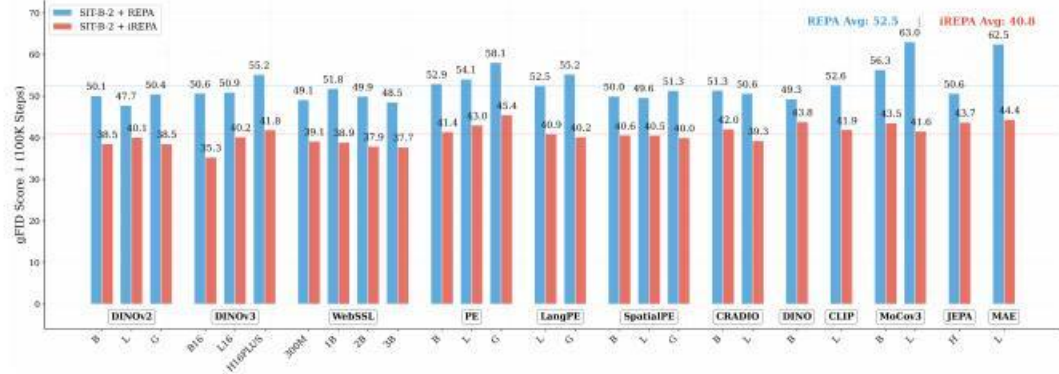
where positive points lie within Manhattan distance $d \leq r_{near}$ from the anchor inside the mask, and negative points lie at distance $d \geq r_{far}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we use a direct contrast metric that measures how much patch-token representations vary across space. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

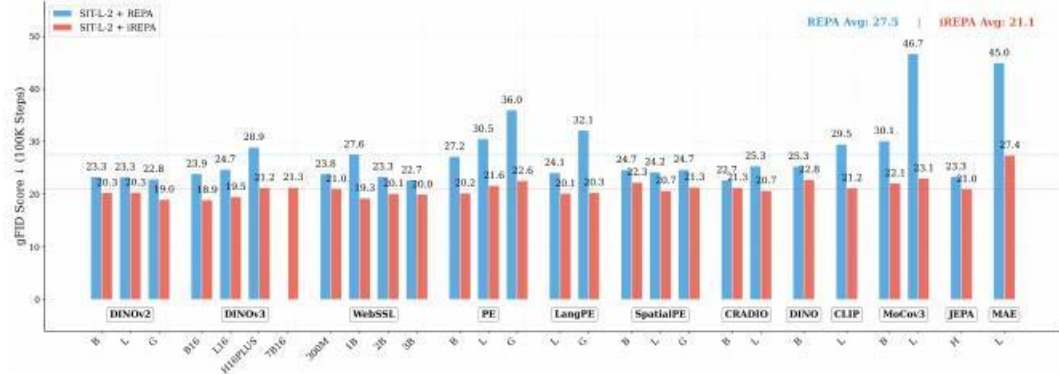
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity and preserved spatial structure, whereas lower values indicate more uniform features and weaker spatial structure, a pattern often associated with global information diminishing spatial contrast (Siméoni et al., 2025).

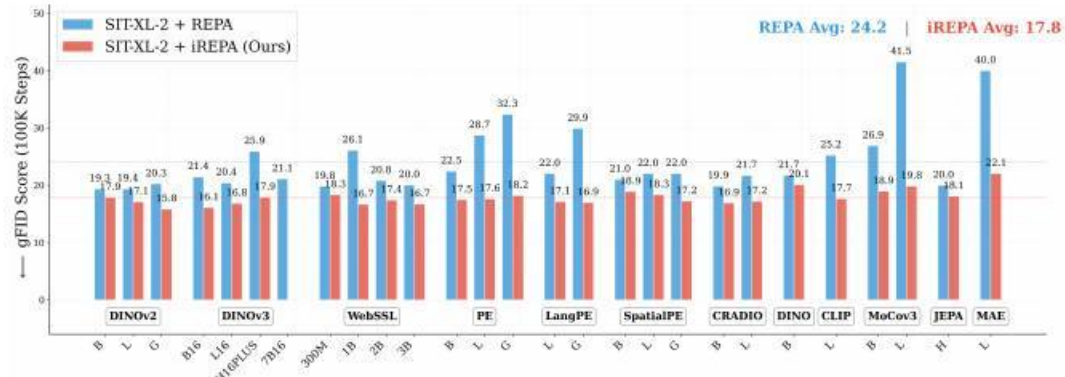
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



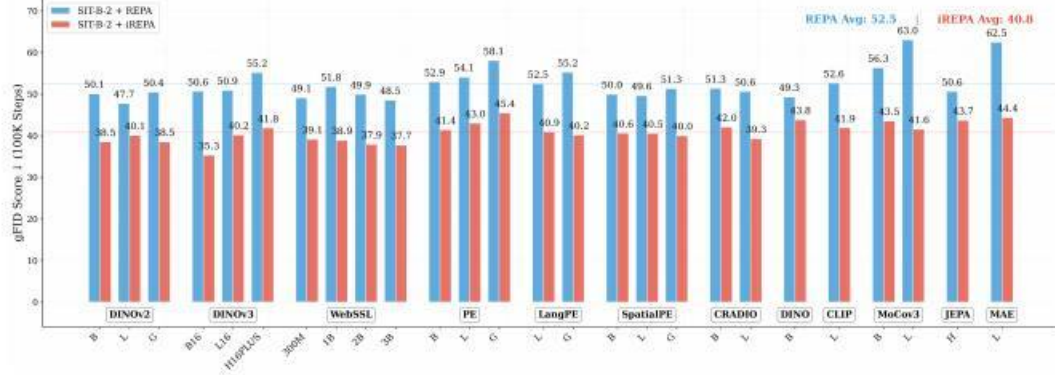
(b) SiT-L-2 (458M parameters)



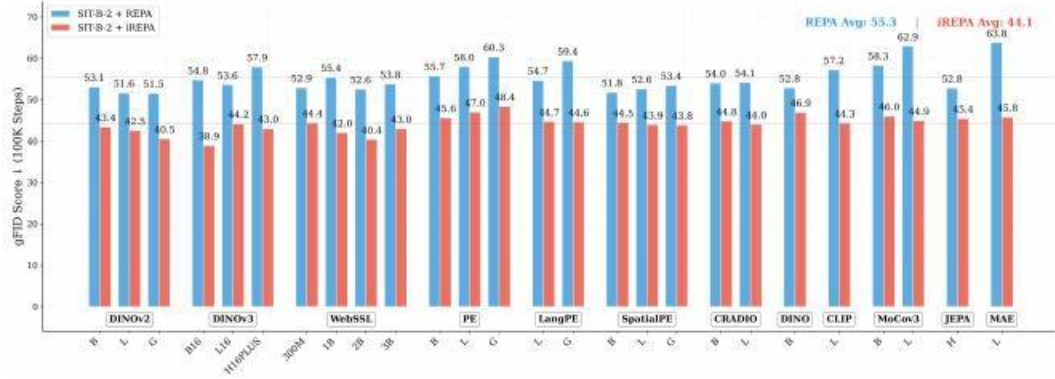
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across vision encoders. iREPA improves performance at both depths, with slightly stronger results for 8 encoder layers. Both configurations show clear FID gains when spatial transfer is strengthened.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across target encoders. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across encoders, with especially strong gains in FID and IS. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

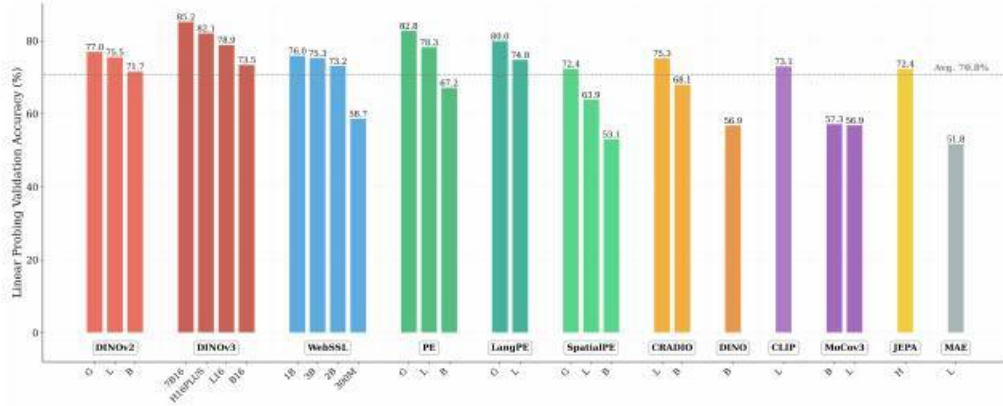


Figure 14: **Global information in mean patch tokens.** Beyond the [CLS] token, the mean of patch tokens contains substantial global semantic information. This global component improves global performance but reduces contrast between individual patch tokens, weakening spatial-structure transfer. Removing part of this information through the patch-token mean strengthens spatial transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA vs iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both IS and FID metrics across all encoders.

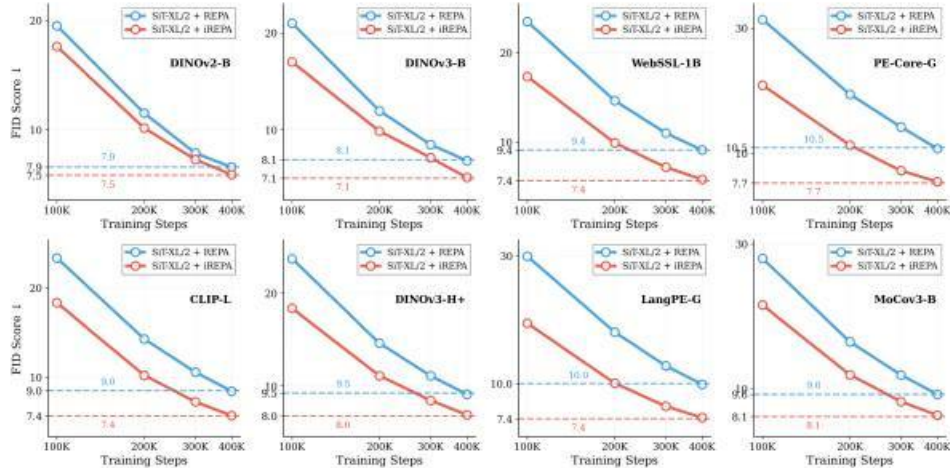


Figure 15: **Accentuating spatial features consistently improves convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

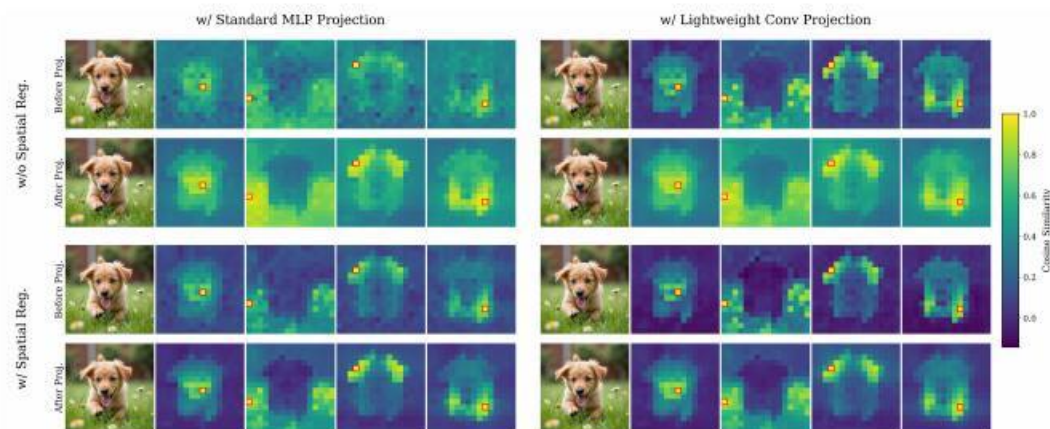


Figure 16: **Visualizing two direct improvements that strengthen spatial feature transfer.** First, the standard MLP projection layer (top-left) loses spatial information while transferring features from the pretrained encoder (after projection) to the diffusion model (before projection). A simpler convolution layer preserves this spatial transfer more effectively (top-right). **Second**, vision-encoder features often have limited spatial contrast (§2). Tokens in one semantic region (*e.g.*, dog) can show high cosine similarity with unrelated tokens (*e.g.*, background). A simple spatial regularization layer accentuates spatial contrast in the learned representation and improves generation performance. The strongest results use both components (Table 2).

G IMPLEMENTATION DETAILS

Training setup. We follow the REPA (Yu et al., 2024) setup unless otherwise specified. All training uses the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol: each image is center-cropped and resized to 256×256 . We use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout diffusion training and inference. For the spatial normalization layer we use $\gamma \in [0.6, 0.8]$, and for the projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` for efficiency, together with gradient clipping and an exponential moving average (EMA) of the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementations. For MeanFlow (Geng et al., 2025), we use the implementation (including REPA) provided by the original authors and add the two iREPA modifications described in §4.

Evaluation. For image-generation evaluation, we follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.), and recall (Rec.) (Kynkäänniemi et al., 2019), all measured on 50K generated images. For sampling, we follow REPA (Yu et al., 2024) and use the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) provide robust local descriptors for tasks ranging from object detection to image matching. Although generative modeling has recently emphasized modern self-supervised representations with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021), our results point to a different selection principle. Because spatial structure is critical for generation quality, classical spatial features can also improve diffusion training when properly aligned. This supports a broader view of useful alignment targets: handcrafted and learned representations are both valuable when they preserve strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders support generative models in several roles: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers for adversarial distillation in diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features improve convergence and enable scaling to large datasets, as shown by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation uses pretrained encoders to guide student networks toward higher-fidelity samples. REPA (Yu et al., 2024) directly aligns diffusion features with external encoders and improves both convergence and quality. We build on this direction by identifying the encoder property that matters most for alignment: preservation of spatial structure rather than discriminative strength alone.

Denosing transformers. Transformer architectures are now central to scalable generative modeling, including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent systems such as GenTron (Chen et al., 2023) scale transformers beyond 3B parameters for text-to-image synthesis, while U-ViT (Bao et al., 2023) shows that pure transformer backbones can compete without U-Net inductive biases. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation, enabling flexible quality-speed trade-offs. Despite these architectural advances, training from scratch remains expensive and often requires millions of iterations. REPA-style alignment methods (Yu et al., 2024) substantially accelerate convergence, and our analysis explains why: the key benefit is the transfer of spatial structure, not merely semantic alignment. This mechanism clarifies why some encoders accelerate training more effectively than others and guides stronger alignment designs. While (Wang

et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as self-supervised learning task. The connection between denoising and representation learning has been studied from several angles. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning and showed that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder can provide strong self-supervised performance. Wang & He (2025) introduce a dispersive loss that encourages internal representation diversity and improves generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA), aligning a diffusion transformer’s latent features across noise levels to provide self-guidance without an auxiliary model. These works establish denoising as a powerful self-supervised task. Our findings complement them by showing that external alignment is most effective when it transfers strong spatial representations, making spatial structure preservation a key factor in the synergy between representation learning and generation.

Scaling self-supervised vision encoders. Self-supervised vision encoders have rapidly scaled in model size and data. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) shows that visual models trained on more than 2 billion images can match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) distills multiple teachers into versatile encoders across visual domains. I-JEPA (Assran et al., 2023) learns by predicting masked regions in representation space, SAM (Ravi et al., 2024) specializes in promptable segmentation, and the Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features can outperform final representations on dense prediction. These models are often judged by global classification-style metrics, but our results show that such metrics do not reliably predict generation quality. For generative alignment, encoders that preserve spatial structure provide the strongest benefit, even when their classification accuracy is lower. This calls for evaluating self-supervised encoders for generative applications with spatial coherence as a first-class criterion.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) to help consider variations of the spatial structure metrics discussed in §3. All figures are generated directly from experiment logs and checkpoints using Claude-Code (Anthropic, 2025). We also use LLM assistance for searching and drafting related-work material in Appendix H, and we use Cursor for limited paper-writing support.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap and interpret increased validation accuracy as evidence for REPA’s effectiveness. Our results show that spatial structure metrics (SSM) provide a stronger view of this gap and explain improvements across models. Figure 17 shows that representation alignment closes the spatial-structure gap between a pretrained encoder such as DINOv2 and the diffusion features.

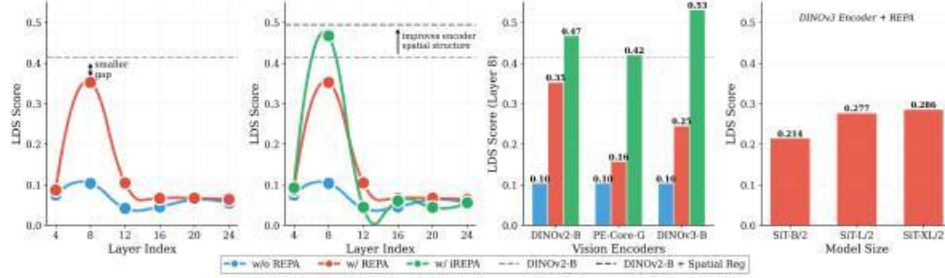


Figure 17: **Representation alignment as bridging the spatial feature gap.** a) Representation alignment closes the spatial feature gap between diffusion features and vision-encoder patch features. b) iREPA (§4) deliberately accentuates vision-encoder spatial features, trading off some global information to improve generation performance. c) iREPA consistently improves performance across vision encoders. d) Spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

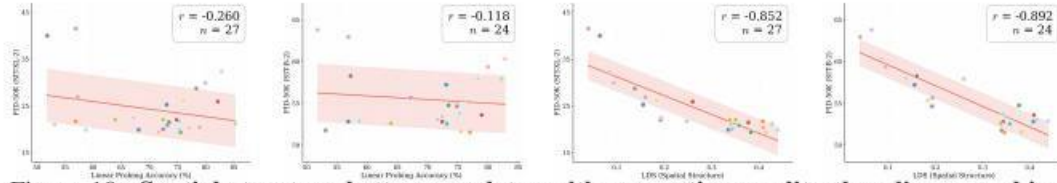


Figure 18: **Spatial structure better correlates with generation quality than linear probing.** Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.260$) and SiT-B-2 ($r = -0.118$) shows weak correlation. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.852$) and SiT-B-2 ($r = -0.892$) shows strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

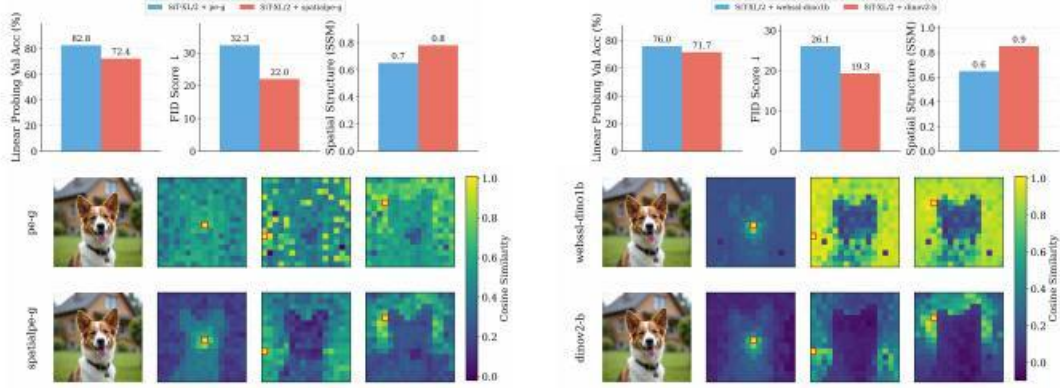


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** The metric comparisons reveal a direct mismatch between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, along with much weaker spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b has higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, again with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine-similarity visualizations confirm the mechanism: encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, whereas classification-oriented encoders (PE-g, WebSSL) lose spatial structure.

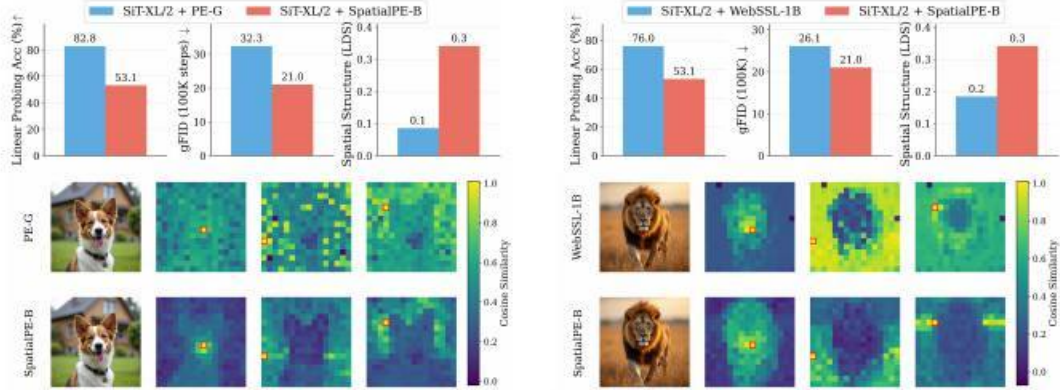


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** Spatial structure provides the stronger predictor of generation quality, outperforming global performance as a selection signal. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize how spatial normalization changes vision-encoder representations. Vision encoders can contain strong global components or overlays that suppress contrast between spatial tokens. For example, tokens in one semantic region can remain highly correlated with tokens in unrelated regions (e.g., background), reducing spatial distinctiveness. Spatial normalization removes this global overlay, strengthens contrast between spatial tokens, and preserves local spatial structure that is more useful for generation.

The following figures show this effect across examples: spatial normalization transforms the feature representations to emphasize spatial structure more clearly.

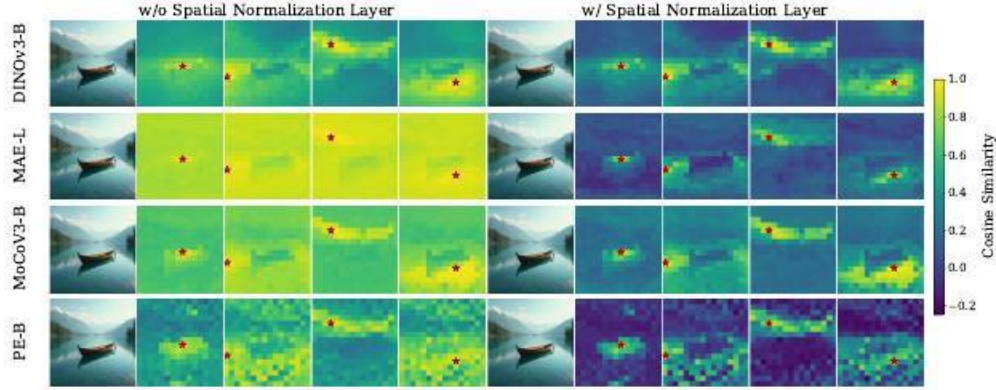


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

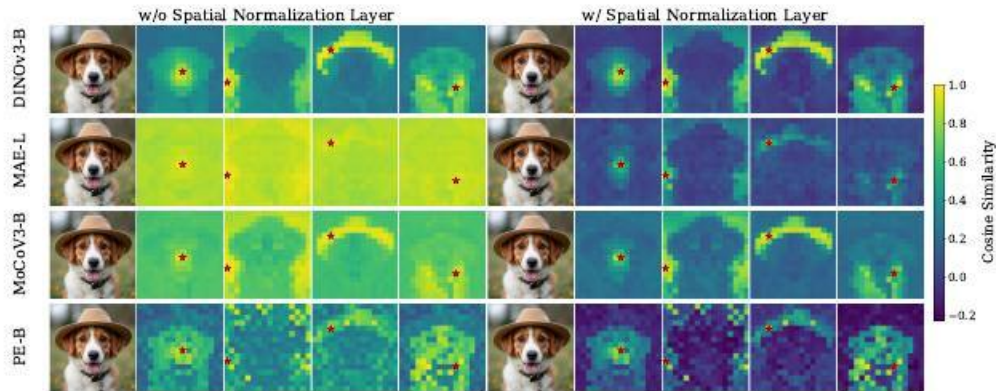


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

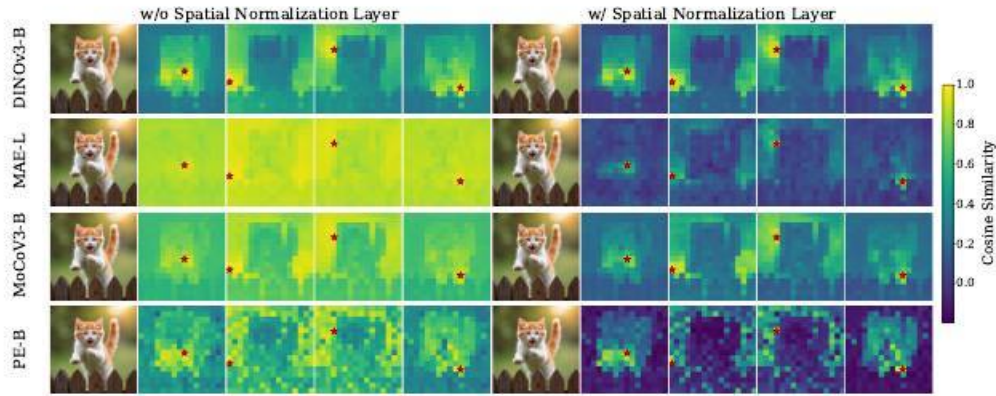


Figure 23: Visualizing impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between different semantic regions.

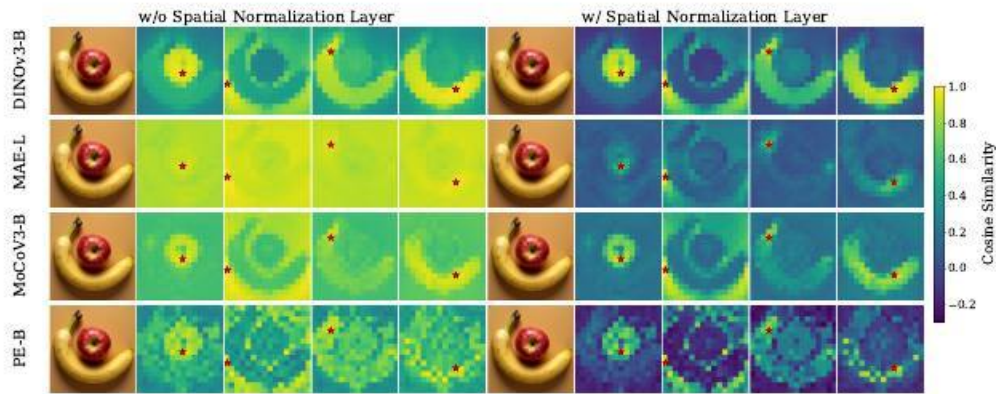


Figure 24: **Visualizing impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We run additional ImageNet-512 (Deng et al., 2009) experiments to test whether the spatial improvements hold at higher resolution. We follow the REPA (Yu et al., 2024) setup and report results across pretrained encoders including DINOv2, DINOv3, WebSSL, and PE. All results use SiT-XL and SiT-B with 50 NFE at inference and no classifier-free guidance. Fig. 25 compares REPA before and after adding iREPA. The spatial improvements generalize to higher resolution and deliver consistent gains across the tested pretrained encoders.

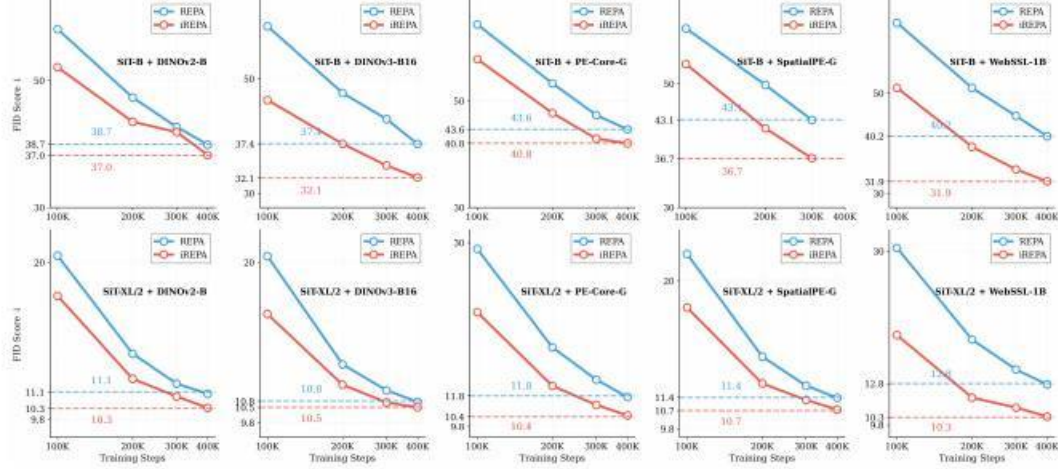


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently improves convergence speed across resolutions, including Imagenet 256 (Figure 7) and Imagenet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To test whether the spatial improvements extend beyond ImageNet, we also run multimodal T2I experiments. Following REPA (Yu et al., 2024), we use MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with pretrained vision encoders including DINOv2, CLIP, WebSSL, and PE. As in REPA (Yu et al., 2024), all models are trained for 150K steps with batch size 256 and evaluated using an ODE sampler with 50 NFE. Fig. 26 compares REPA before and after adding iREPA. The spatial improvements generalize to multimodal T2I and yield consistent gains across the tested pretrained encoders.

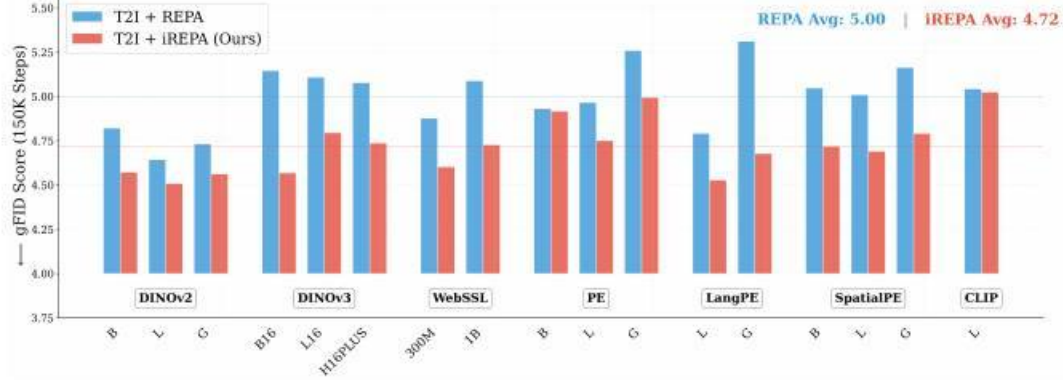


Figure 26: **Text-to-image generation across encoder variants.** Accentuating spatial-feature transfer from target representations to diffusion features consistently improves convergence speed for both ImageNet (see §4) and multimodal T2I tasks (above). The gains hold across pretrained encoders including DINOv2, CLIP, WebSSL, and PE.

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, the identified outliers. Figure 28 reports the updated correlations across model sizes. Spatial structure remains much more correlated with generation performance than linear probing accuracy. After outlier removal, linear probing even shows a small positive correlation with gFID, meaning that higher linear probing accuracy corresponds to worse generation in this subset. This trend reinforces the observation in Sec. 2: target representations with stronger global semantic performance can yield similar or worse REPA performance.

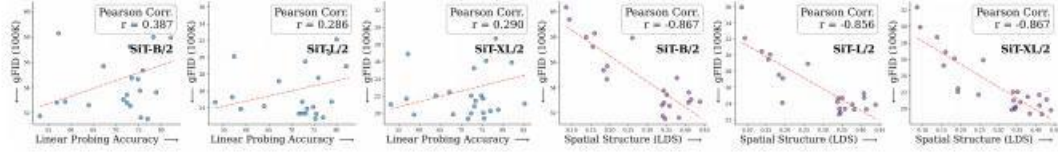


Figure 27: Correlation analysis without outliers. Across model sizes (B, L, XL), spatial structure remains much more correlated with generation performance (Pearson $|r| > 0.85$) than linear probing accuracy (Pearson $|r| < 0.38$) after removing MoCOv3-L and MAE-L. After removing these outliers, linear probing shows a small positive correlation with gFID, indicating that higher linear probing accuracy can correspond to worse generation. This trend matches Sec. 2: target representations with stronger global semantic performance can produce similar or worse REPA performance.

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after applying spatial improvements to REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and classifier-free guidance scale 4.0. Consistent with the quantitative results (§4), iREPA improves the visual quality and coherence of generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we also verify the robustness of the proposed findings with alternative evaluation metrics for generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we use CMMD (Jayasumana et al., 2024) with PyTorch implementation from (Paul, 2024). We use the standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). We check whether key findings from §3 and §4 remain consistent when using CMMD.

Spatial structure correlates much higher with generation performance than linear probing. To test robustness of the analysis in §3, we repeat the large-scale encoder correlation study using CMMD (Jayasumana et al., 2024). Figure 29 reports results for SiT-B/2 (100K steps) with REPA. All spatial metrics correlate much more strongly with CMMD (Pearson $|r| > 0.88$) than linear probing does (Pearson $|r| = 0.074$), confirming that the key empirical findings from §3 are not specific to FID.

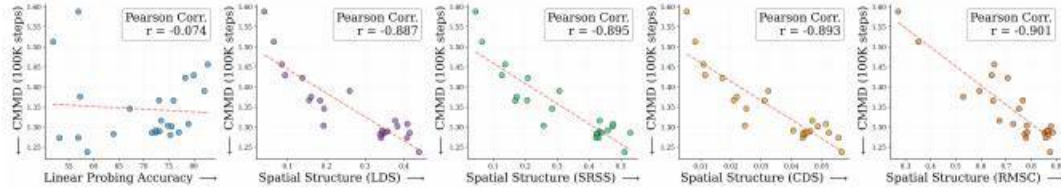


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 using CMMD (Jayasumana et al., 2024) instead of gFID. All results use SiT-B/2 (100K steps) and REPA. Every spatial metric correlates much more strongly with CMMD (Pearson $|r| > 0.88$) than linear probing does (Pearson $|r| = 0.074$). This demonstrates that key empirical findings from §3 are robust to the evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We also evaluate the experimental findings from §4 using CMMD (Jayasumana et al., 2024) alongside standard metrics (gFID, sFID, IS, etc.). Table 8 shows that iREPA improves convergence across vision encoders under both CMMD and the traditional metrics (IS, FID, sFID, Prec., Recall). The gains hold with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We report additional results with CMMD (Jayasumana et al., 2024). All results use SiT-XL/2, batch size 256, and traditional REPA as the baseline. We adopt the PyTorch implementation from (Paul, 2024) and use the standard reference images from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). Across vision encoders, accentuating spatial-feature transfer with iREPA improves convergence under both CMMD and traditional metrics (IS, FID, etc.). These results confirm that the empirical gains in §4 are robust to the evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 evaluate spatial normalization (§4) on SAM2-S (46M) encoder features². Table 9 shows that for SAM2, spatial normalization improves performance, but the gains are smaller than for encoders with stronger global components. This matches the mechanism: spatial normalization removes the global component (mean of patch tokens) to enhance spatial contrast, and SAM2 already contains little global information (validation accuracy < 24%). Consequently, spatial normalization only modestly increases spatial contrast for SAM2 (see Figure 30).

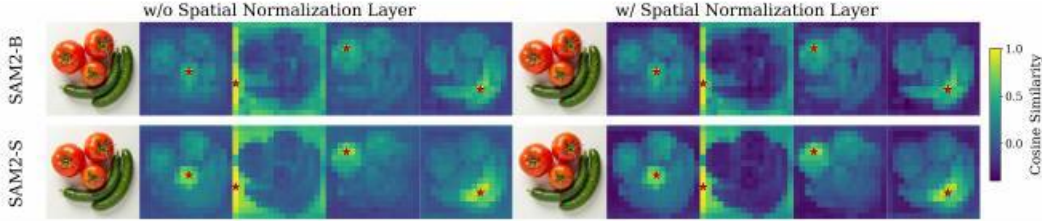


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, spatial normalization enhances spatial contrast, but the gains are naturally smaller. Spatial normalization (§4) works by removing the global component (mean of patch tokens) to sharpen spatial contrast. Because SAM2 already has little global information (validation accuracy < 24%), the layer only modestly increases spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** iREPA improves SAM2-S despite SAM2 already having limited global information. All results use SiT-XL/2 without classifier-free guidance, SAM2-S (46M) as the vision encoder, and traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full-finetuning³ instead of linear probing. Fig. 31 reports the results. All results use SiT-XL/2 (100K steps) and REPA. To obtain full-finetuning validation accuracy, we finetune each vision encoder with learning rate $5e-5$, warmup ratio 0.1, and 30 total epochs. Spatial metrics remain much more correlated with generation performance than full-finetuning accuracy. Notably, gFID has a weak positive correlation with validation accuracy after full-finetuning (Pearson $r = 0.317$), meaning that higher validation accuracy can coincide with worse generation. This is consistent with §2: target representations with higher global semantic performance often produce similar or worse REPA performance.

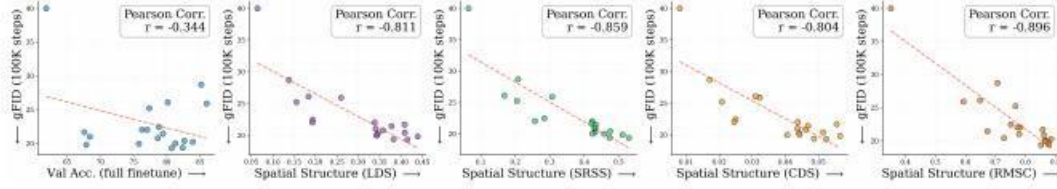


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full-finetuning instead of linear probing. All results use SiT-XL/2 (100K steps) and REPA. Spatial metrics remain much more correlated with generation performance than full-finetuning accuracy. gFID has a weak positive correlation with validation accuracy after full-finetuning (Pearson $r = 0.317$), meaning that higher validation accuracy can coincide with worse generation. This matches §2: target representations with higher global semantic performance often produce similar or worse REPA performance.

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recently proposed state-of-the-art pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results use 200K training iterations, batch size 256, and 50-step Euler sampling without classifier-free guidance. DeCo results are taken directly from (Ma et al., 2025b).